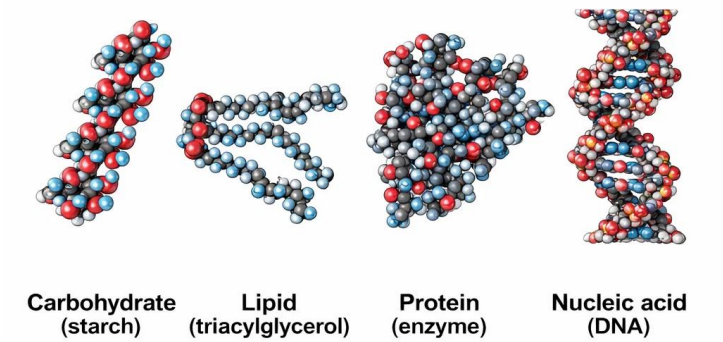
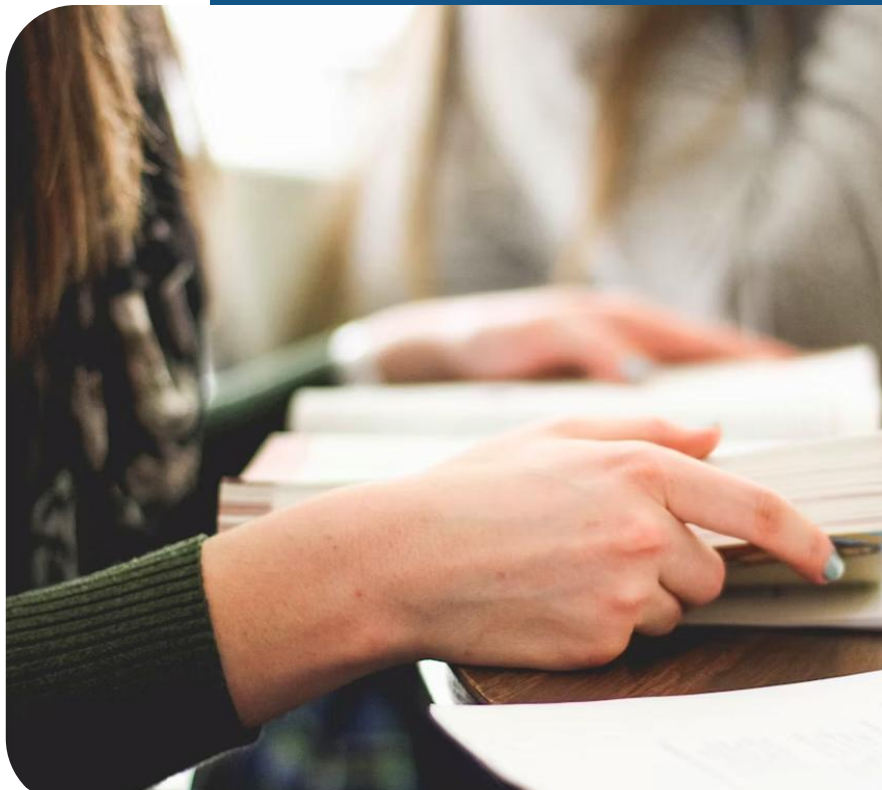
Biological Macromolecules



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Protein



STRUCTURE & FUNCTIONS OF PROTEINS

STRUCTURE

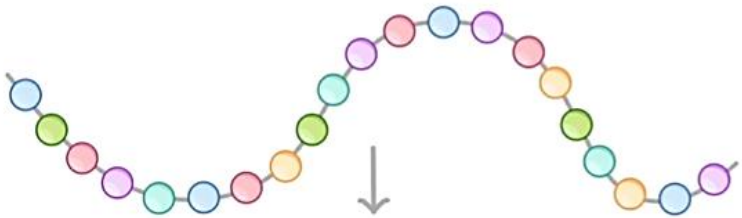
MADE UP OF LOTS OF SMALL MONOMERS:
(AMINO ACIDS)



2 AMINO ACIDS JOIN TO FORM A DIMER:
(DIPEPTIDE)



3+ AMINO ACIDS LINKED TOGETHER FORM A POLYMER:
(POLYPEPTIDE)



POLYPEPTIDES CAN COIL, FOLD & JOIN TOGETHER
TO MAKE PROTEINS



FUNCTIONS

1. ACT AS **ENZYMES**
   - USED TO BREAK DOWN & BUILD UP OTHER MOLECULES



2. FORM **ANTIBODIES**
   - CRUCIAL PART OF IMMUNE RESPONSE



3. HELP **TRANSPORT** MOLECULES & IONS ACROSS CELL MEMBRANES



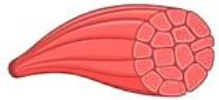
4. **STRUCTURAL COMPONENTS**
   - CREATE STRONG FIBRES (KERATIN & COLLAGEN)



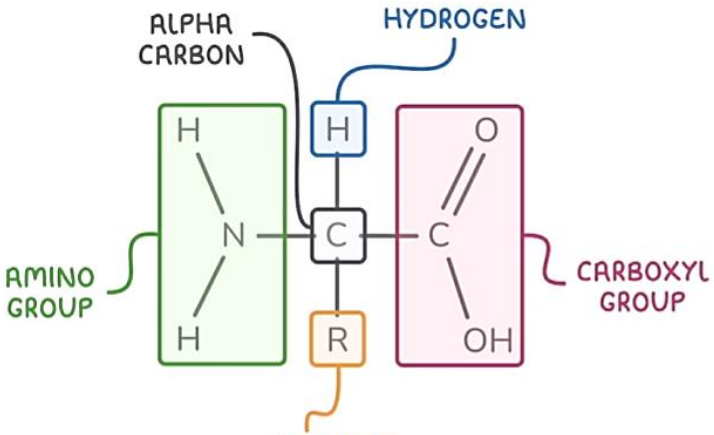
5. FUNCTION AS **HORMONES**
   - CHEMICAL MESSENGERS



6. **MUSCLES** ARE MADE OF PROTEINS
   - ALLOW MUSCLE CONTRACTION

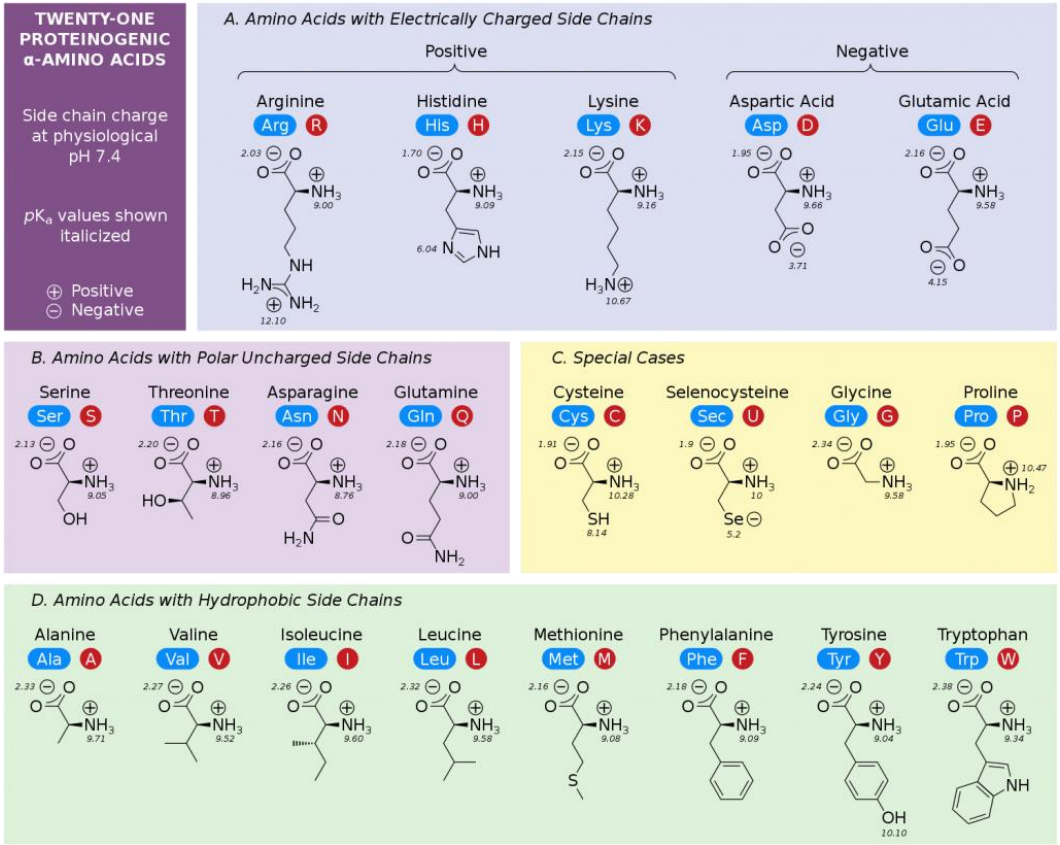


STRUCTURE OF AMINO ACIDS



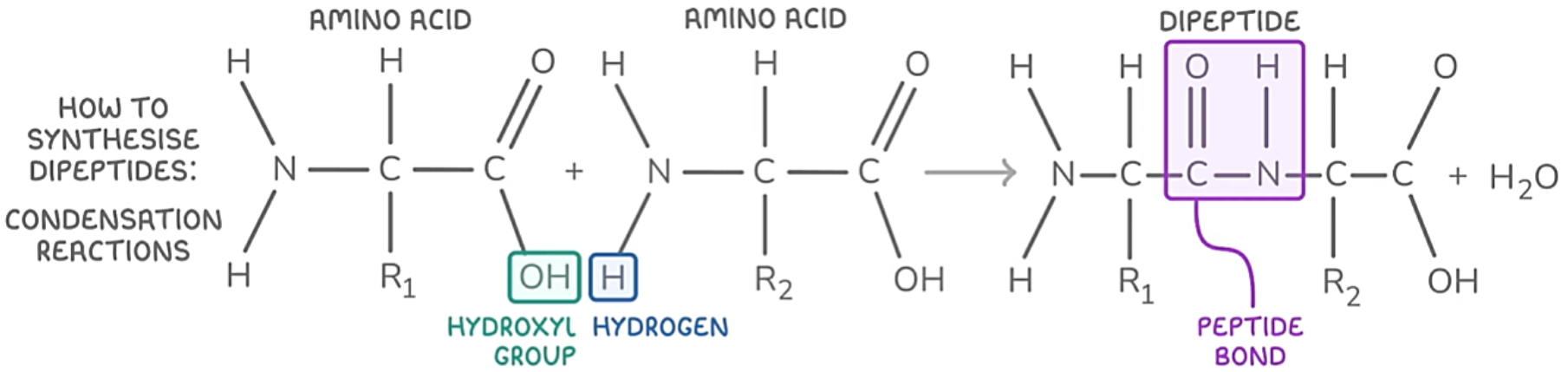


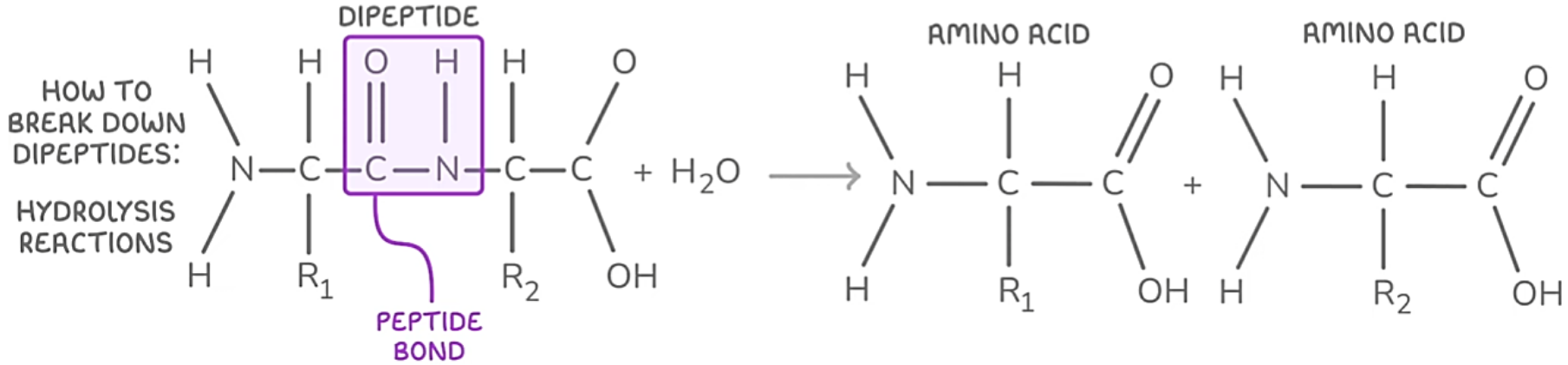
20 DIFFERENT AMINO ACIDS



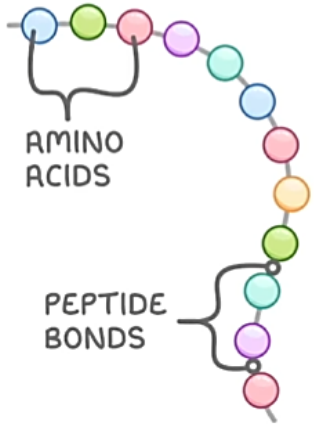
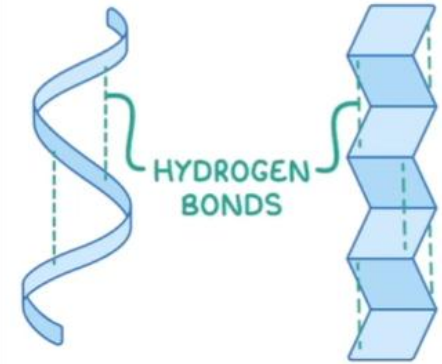
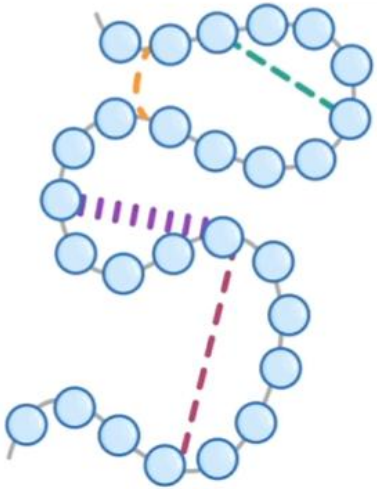
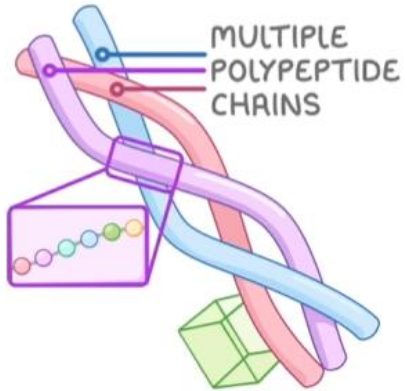
*selenocysteine is not part of the 20 essential amino acids that are in proteins.

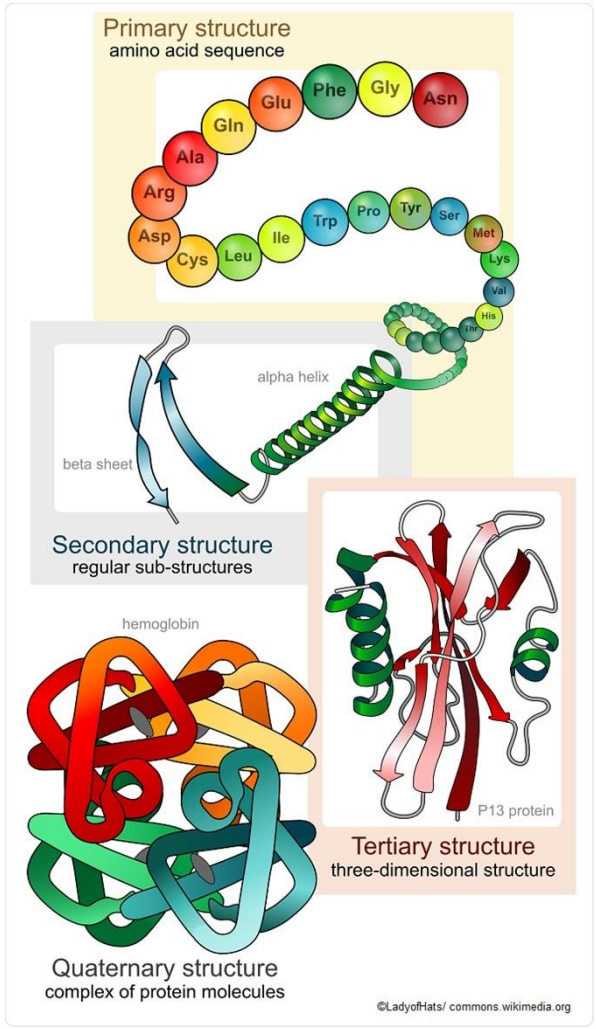
HOW DIPEPTIDES ARE SYNTHESISED & BROKEN DOWN

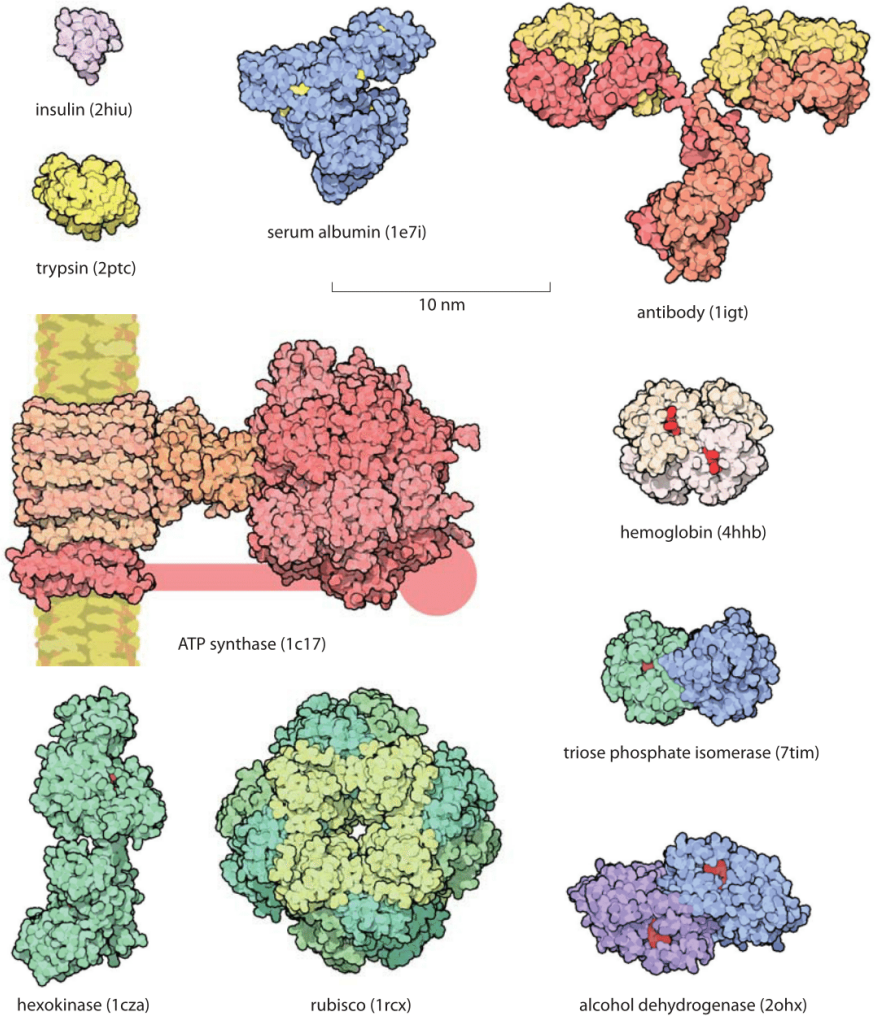




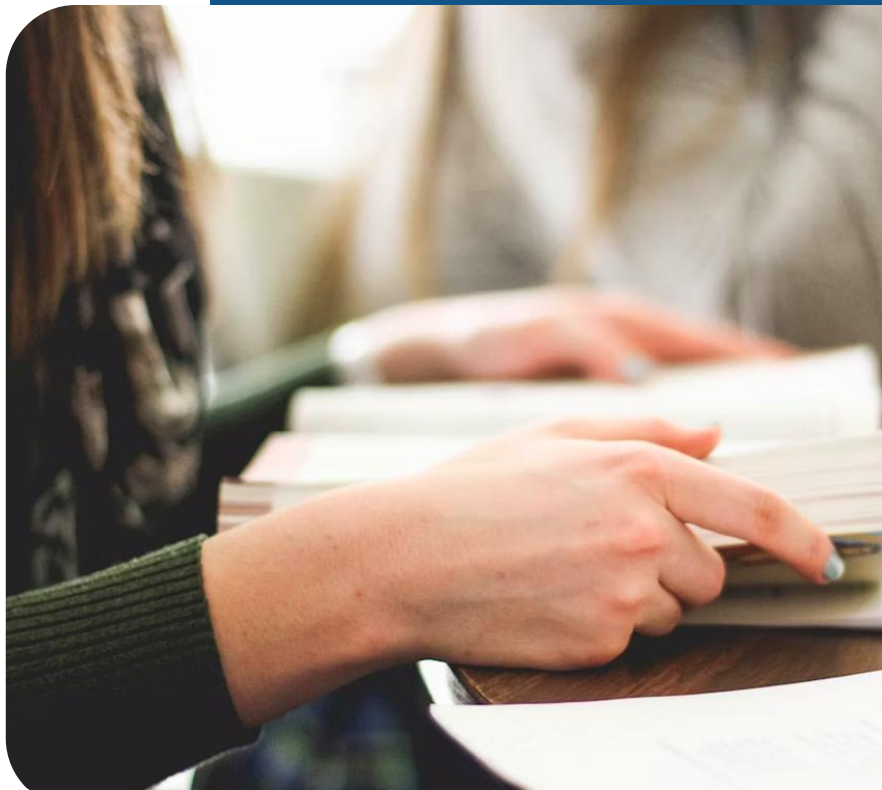
LEVELS OF PROTEIN STRUCTURE

	PRIMARY STRUCTURE	SECONDARY STRUCTURE	TERTIARY STRUCTURE	(ISN'T ALWAYS PRESENT) QUATERNARY STRUCTURE
WHAT IT MEANS	• SEQUENCE OF AMINO ACIDS	• INTERACTIONS BETWEEN AMINO ACIDS CAUSE CHAIN TO COIL/FOLD	• CHAIN FOLDS & TWISTS INTO UNIQUE 3D SHAPE	• 2+ POLYPEPTIDE CHAINS JOIN TOGETHER
TYPES OF BONDS	• PEPTIDE BONDS	• HYDROGEN BONDS (BETWEEN AMINO GROUP & CARBOXYL GROUP OF 2 AMINO ACIDS)	• HYDROGEN BONDS (STRONG COLLECTIVELY) • IONIC BONDS (POSITIVE & NEGATIVE R GROUPS) • DISULFIDE BRIDGES (SULFUR-CONTAINING R GROUPS) • HYDROPHOBIC & HYDROPHILIC INTERACTIONS (POLAR & NON-POLAR R GROUPS)	
SHAPE	 POLYPEPTIDE CHAIN	 REPETITIVE & PREDICTABLE	 GIVES SPECIFIC FUNCTION	 MAY INCLUDE PROSTHETIC GROUPS (NON-PROTEIN GROUPS)

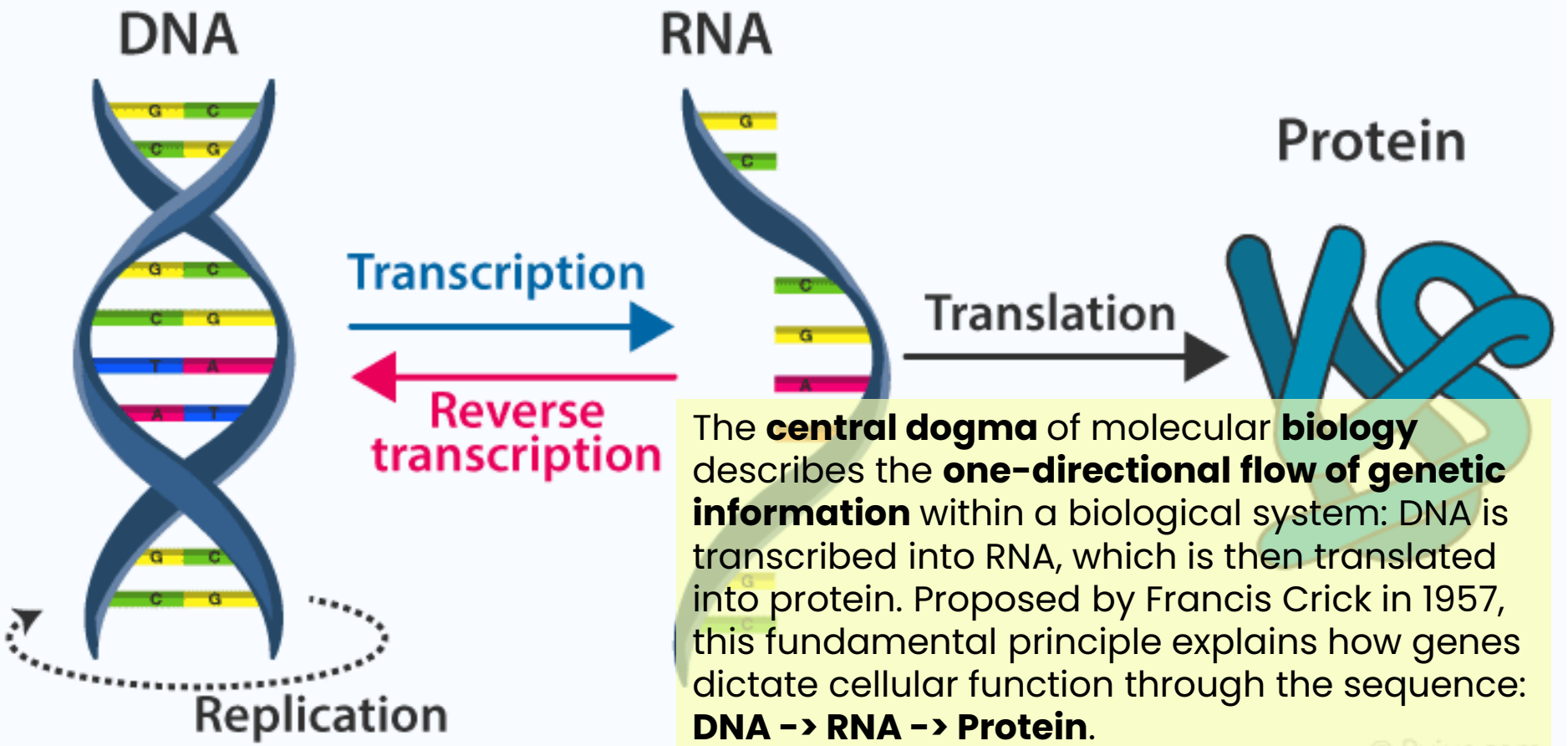

Primary structure
amino acid sequence
Glu
Phe
Gly
Asn
Gln
Ala
Arg
Asp
Cys
Leu
Ile
Trp
Pro
Tyr
Ser
Met
Lys
Val
His
Thr
alpha helix
beta sheet
Secondary structure
regular sub-structures
hemoglobin
P13 protein
Tertiary structure
three-dimensional structure
Quaternary structure
complex of protein molecules
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insulin (2hiu)
trypsin (2ptc)
serum albumin (1e7i)
10 nm
antibody (1igt)
ATP synthase (1c17)
hemoglobin (4hhb)
triose phosphate isomerase (7tim)
hexokinase (1cza)
rubisco (1rcx)
alcohol dehydrogenase (2ohx)

Central Dogma of Biology



CENTRAL DOGMA : DNA TO RNA TO PROTEIN



The **central dogma** of molecular **biology** describes the **one-directional flow of genetic information** within a biological system: DNA is transcribed into RNA, which is then translated into protein. Proposed by Francis Crick in 1957, this fundamental principle explains how genes dictate cellular function through the sequence: **DNA –> RNA –> Protein**.

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